

Objective: Predict 24-hour ahead electricity demand for the specified region (Bareilly/Mathura).

Data Preparation:

- 1. Data Coverage: Meter readings from 2020/2021 (Bareilly) and 2019/2020 (Mathura) were combined to ensure robust, multi-year training history.
- 2. Resampling: Raw 3-minute readings were aggregated (summed) into 1-hour intervals, establishing the prediction frequency.
- 3. Imputation: Missing values were handled by forward-filling up to 3 hours, followed by linear interpolation for remaining gaps.
- 4. Outliers: Data stability was ensured by clipping extreme consumption values at the 1st and 99th historical percentiles.

Modeling Approach:

- 1. Baseline: Seasonal Naive, which predicts consumption for a future hour based on the consumption from the same hour on the previous day.
- 2. HistGradientBoostingRegressor Model: A model trained on a 7-day rolling historical window (ending at origin ime T).
 - Features: Lag consumption (1, 2, 3 hours), 24h rolling mean, day-of-week, and cyclical Sin/Cos of hour-of-day. Temperature is included if external API fetch is successful.
 - Forecasting: Predictions were generated iteratively, where the forecast for $T + h$ is used to derive features (lags) needed for the $T + h + 1$ prediction.

Performance Metrics:

Model	MAE	WAPE	sMAPE	RMSE
HGBoost	0.048	0.349	0.360	0.066
Seasonal Naive	0.062	0.452	0.464	0.087

